Data Structures

- Linear
  - Array
    - Static Array
      - class Array
    - Dynamic Array
      - class ArrayList
      - class List<T>
  - Stack
    - Array-based Stack
      - class Stack<T>
    - Linked List-based Stack
  - Queue
    - Simple Queue
      - class Queue<T>
    - Deque (Double-Ended Queue)
      - class LinkedList<T>
    - Priority Queue
    - Circular Queue
  - Linked List
    - Singly Linked List
    - Doubly Linked List
      - class LinkedList<T>
    - Circular Linked List
    - Skip List
- Tree
  - Binary Tree
    - By Structural Property
      - Full Binary Tree
      - Complete Binary Tree
      - Perfect Binary Tree
    - By Functional Purpose
      - Binary Search Tree (BST)
        - Balanced BST
          - AVL Tree
          - Red-Black Tree
        - Unbalanced BST
      - Heap
        - Min-Heap
        - Max-Heap
      - Huffman Tree
      - Segment Tree

            - KD-Tree
    - Multiway Tree
        - Multiway Search Trees
            - B-Tree
            - B+ Tree
            - 2-3 Tree
            - 2-3-4 Tree
        - Trie Tree
            - Standard Trie
            - Compressed Trie
            - Suffix Tree
        - Space Partitioning Trees
            - Quadtree
            - Octree

- Graph
    - Directed / Undirected
    - Weighted / Unweighted
    - Representation
        - Adjacency Matrix
        - Adjacency List
        - Adjacency Multilist
        - Orthogonal List
    - Special Graphs
        - DAG
        - Tree Graph
        - Bipartite

- Hash
    - class Hashtable
    - class HashSet<T>
    - class Dictionary<TKey,TValue>
    - class OrderedDictionary

- Specialized / Utility
    - Sorted
        - class SortedSet<T>
        - class SortedDictionary<TKey,TValue>
        - class SortedList
        - class SortedList<TKey,TValue>
    - ReadOnly
        - class ReadOnlyCollection<T>
        - class ReadOnlyDictionary<TKey,TValue>
        - interface IReadOnlyList<T>
        - interface IReadOnlyDictionary<TKey,TValue>
    - Data Binding

        - class ObservableCollection<T>
    - String
        - class StringCollection
        - class StringDictionary
    - Bit Operation
        - class BitArray
        - class BitVector32
    - Hybrid
        - class HybridDictionary
        - class NameValueCollection
    - Immutable
        - class ImmutableArray<T>
        - class ImmutableList<T>
        - class ImmutableHashSet<T>
        - class ImmutableDictionary<TKey,TValue>
        - class ImmutableSortedDictionary<TKey,TValue>
    - Disjoint Set / Union-Find
    - Bloom Filter
    - Counting Filter
    - Linear Probing Table

- Concurrent

    - class ConcurrentQueue<T>
    - class ConcurrentStack<T>
    - class ConcurrentBag<T>
    - class ConcurrentDictionary<TKey,TValue>
    - class BlockingCollection<T>